THE ADEQUACY OF PHYSICIAN SUPPLY IN SMALL RURAL COUNTIES

by David A. Kindig and Hormoz Movassaghi

Prologue: *Roughly one-quarter of Americans live in rural areas. Rural Americans share the same basic needs as do urban dwellers, one of the most fundamental of which is the need for health care. There is growing attention on the rural health scene, including the health status of rural residents, the status of rural hospitals, and the numbers of physicians that choose to practice in rural areas. In this article, David Kindig and Hormoz Movassaghi examine physician supply in small rural counties—those having fewer than 10,000 residents. Over the period 1975–1985, they find, the number of physicians per 100,000 residents grew almost three times faster for the entire United States than for rural areas, despite overall growth in supply. "To consider this [issue]," they state, "we must have some standard or norm of availability that is desired or optimal for these locations. To our knowledge, no such standard exists." Kindig, who received his medical degree and a doctorate in experimental pathology from the University of Chicago, is professor of preventive medicine and director of programs in health management at the University of Wisconsin, Madison, School of Medicine. His interest in physician supply began when he was president of the American Medical Student Association in the late 1960s, during a time when issues of distribution of physicians across the country first reached the national consciousness. As deputy director of the Bureau of Health Manpower, under the Department of Health, Education and Welfare, from 1974 to 1976, he helped to develop legislation that led to the formation of the Graduate Medical Education National Advisory Committee (GMENAC) and its landmark findings. Movassaghi is currently an assistant professor of finance at the School of Business, Ithaca College, Ithaca, New York. He holds a master of business administration degree and a master's degree in health administration from the University of Wisconsin, Madison, where he was a research assistant at the time this paper was completed.*

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The supply and distribution of physicians in rural areas has been an important area of investigation and public policy over the past two decades. Numerous studies have described the availability of physicians in rural areas, their characteristics, and what motivates them to locate and practice in rural America.[1]

Two seminal reports in the early 1980s were particularly influential in changing the nature of the debate on the issues of physician supply and distribution in rural areas. The first was the Graduate Medical Education National Advisory Committee (GMENAC) report, which indicated that the explosive growth in the output from American medical schools in the 1970s would result in a "surplus" of 70,000 physicians in 1990 and 145,000 in 2000.[2] Notwithstanding some of the criticisms of GMENAC's methodology that have since emerged, there is a consensus that the basic trends predicted are correct; indeed, the growth in overall physician availability may not reach its peak until early in the twenty-first century.[3] Following the GMENAC report, the other major studies of this period were those coming out of The RAND Corporation by William Schwartz, Joseph Newhouse, and their colleagues.[4] In these critical papers, the authors looked at changes in the supply of a selected group of board-certified physicians initially and then all nonfederal physicians in towns with populations of 2,500 to 50,000 or more in twenty-three states during 1960–1977 and 1970–1979, respectively. Their basic finding was that by 1979 nearly every town with more than 2,500 residents had ready access to a physician. They suggested that as the supply of physicians increases in the 1980s, a wider range of health services will become available to populations in nonmetropolitan areas. While the authors carefully qualified their conclusions, the widespread interpretation made from these studies was that the overall increase in physician supply will correct imbalances in the geographic distribution of physicians over time.

Studies by The RAND Corporation employed several sophisticated methodologies such as choosing towns rather than counties as the unit of analysis, an approach that has not been used by most other studies because of its complexity and expense. A number of limitations in the RAND studies, however, figured in our decision to conduct the current investigation. First, basic findings were reported in terms of the percentage of towns in different population groups that had at least one physician, not how many physicians in total were present vis-a-vis the population in each grouping of towns. Furthermore, towns with under 2,500 residents were not included in the RAND studies, and results were dominated by findings in towns with more than 10,000 residents. For example, in towns between 2,500 and 5,000, RAND reported a growth in internists from 2 percent to 9 percent between 1966 and 1977 and similar

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growth for pediatricians from 1 percent to 3 percent. Second, they did not consider the contributions of doctors of osteopathy (DOs) and federal physicians. Third, only twenty-three states were selected from four regional clusters, which may have biased the results. Finally, only limited regional differences were explored with no significant results.

A number of researchers have addressed and pointed out some of the limitations of these studies.[5] Most important, perhaps, is a federal study on the diffusion of primary care physicians.[6] This study confirmed that the general process of diffusion was taking place, yet its projections indicated that even in 1994 and after taking into account the effects of diffusion, 1,148 primary care physicians would still be needed to meet the current federal physician minimum levels in nonmetropolitan areas and 2,056 in metropolitan areas.

We undertook the study on which this article is based to examine changes in physician availability from 1975 to 1985 in counties with fewer than 10,000 inhabitants. We also look at changes in physician supply in counties of low population density ("frontier areas"); contribution of DOs in provision of medical services; and age, specialty, and possible state or regional variations in physician availability. Finally, we compare the current availability of physicians in rural counties to corresponding national figures, and we comment on the relationship of current physician levels to the adequacy of physician availability for those counties.

Trends In Physician Availability

The data used in this study were compiled from 1975 and 1985 Bureau of the Census Current Population Reports, American Medical Association (AMA) Physician Masterfile tapes for 1975 and 1985, and the Yearbook and Directory of the American Osteopathic Association for 1975 and 1985.[7] The 684 counties with less than 10,000 population in both years were divided by county population size (less than 2,500, 2,500–4,999, and 5,000–9,999) and population density (frontier areas contain fewer than six persons per square mile, and nonfrontier areas, more than six). These counties had a 1985 population of approximately four million. Physicians include all active federal and nonfederal MDs and DOs except interns and residents. Primary care physicians refer to those with the self-reported primary specialties of either family practice, general practice, general internal medicine, general pediatrics, or obstetrics/gynecology. Comparative figures for the whole United States and metropolitan counties do not include the forty-two counties designated as "potential standard metropolitan statistical areas (SMSAs)."

Rates of increase. Exhibit 1 illustrates the changes in physicians per

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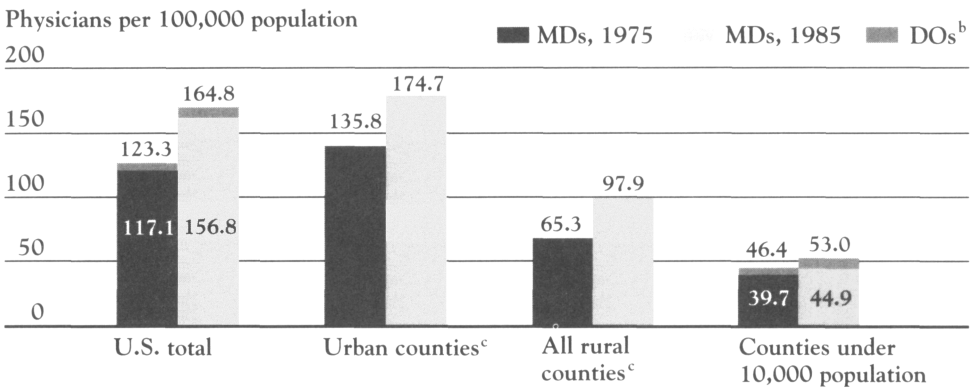
Exhibit 1
Availability Of Patient Care Physicians, 1975–1985[a]



[a] Excludes residents and fellows.
[b] Doctors of osteopathy.
[c] Potential standard metropolitan statistical area (SMSA) counties not included; osteopathic information not available.

100,000 population in four categories of counties between 1975 and 1985. In counties with fewer than 10,000 residents, the ratio increased from 46.4 to 53.0, an increase of 14.2 percent. During the same period, the physician-to-population ratio for the United States as a whole grew from 123.3 to 164.8, a growth rate of 33.6 percent. These two ratios include both allopathic and ostoeopathic physicians, with osteopathic physicians making up 15.3 percent and 5.1 percent of total physicians in counties of under 10,000 population and the whole United States, respectively, in 1985. For comparison, all nonmetropolitan areas grew 48.6 percent from 65.3 to 97.9. These latter ratios refer to allopathic physicians only, so the levels of total physician availability would be somewhat greater than displayed. Over the 1975–1985 period, the mean level of physician availability for the entire United States grew from 2.7 to 3.1 times that in counties of under 10,000 residents despite the growth in supply in both areas.

Exhibit 2 presents the distribution of all physicians who were practicing in counties with a population of less than 10,000 in 1985 and the percentage change over the decade. The absolute number increased from 1,758 in 1975 to 2,110 in 1985, an increase of 20 percent. When these counties are divided by population size, availability is approximately the same for counties larger than 2,500, but the growth rate is greater in the 2,500–4,999 population category. When the same counties are divided into frontier and nonfrontier areas, the less densely populated counties show both a slightly higher availability and a substantially higher rate of growth than those with more than six persons per square mile. This

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Exhibit 2
Physician Availability In 684 U.S. Counties With Fewer Than 10,000 Residents, 1985

Population size	Population density
Less than 2,500 29.9 physicians per 100,000 (13.7%)	Fewer than six persons per square mile 57.4 physicians per 100,000 (18.6%)
2,500–4,999 52.8 physicians per 100,000 (26.9%)	More than six persons per square mile 51.3 physicians per 100,000 (12.3%)
5,000–9,999 54.3 physicians per 100,000 (11.7%)	

Note: Numbers in parentheses denote changes in physician-to-population ratio between 1975 and 1985. Includes allopathic and osteopathic physicians.

counterintuitive finding could possibly be the result of larger population centers in the frontier areas and a more diffuse nonfrontier population; this possibility deserves further examination.

Primary care physicians. There were 1,418 primary care physicians reported to be practicing in counties with resident populations under 10,000 in 1975; by 1985, this number had increased to 1,624. Relative to the population of those counties, however, the ratio of primary care physicians to population grew from 37.3 to 40.8, an increase of 9.4 percent over a decade (Exhibit 3). This was nearly one-third or less of the similar growth rates in the entire United States (30 percent), all metropolitan

Exhibit 3
Availability Of Primary Care Physicians, 1975–1985[a]

	1975		1985	
	Physicians per 100,000	Percent of patient care physicians	Physicians per 100,000	Percent of patient care physicians
U.S. total[b]	53.9	46.0%	70.4	44.9%
Metropolitan counties[b]	59.5	43.8	75.5	43.2
All nonmetropolitan counties[b]	38.1	58.4	53.9	55.0
Counties with less than 10,000 population[b]	32.2	81.0	35.0	77.9
Counties with less than 10,000 population[c]	37.3	80.7	40.8	77.0

[a] Excludes residents and fellows; includes general practice, family practice, general internal medicine, general pediatrics, and obstetrics/gynecology.
[b] Allopathic physicians only.
[c] Includes osteopathic physicians.

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areas (27 percent), and nonmetropolitan areas (39 percent).

As a component of patient care physicians, primary care physicians constituted a much higher percentage of all physicians in small rural counties than either the United States, metropolitan, or nonmetropolitan areas over the period of the study. In 1985, 77 percent of patient care physicians in rural areas were in primary care specialties, while the corresponding percentages in the United States, metropolitan, and nonmetropolitan areas were 45 percent, 43 percent, and 55 percent, respectively. Within the small rural counties, frontier areas had a higher number of primary care physicians relative to their population than nonfrontier areas, both in 1975 and 1985. In fact, over the period of this study, the former seem to have experienced a faster growth rate of primary care physicians than the latter. Within the subspecialties of primary care, family and general practitioners accounted for 91 percent of all primary care physicians practicing in those small rural counties in 1985. There was no significant difference in this respect between the frontier and nonfrontier areas or between the three population categories.

Board certification. The number of board-certified physicians, particularly those in primary care specialties, clearly increased. In 1975, only 18 percent of physicians practicing in small rural counties were board certified (15 percent of primary care and 34 percent of nonprimary care physicians). By 1985, that percentage had climbed to 50 percent and was distributed evenly among physicians in primary care and other specialties. The majority of DOs practicing in these areas were not board certified. The proportion of DOs that were board certified increased from 4.8 percent in 1975 to 9.4 percent in 1985.

Demographic characteristics. We examined separately the physicians under age thirty-five and those over age sixty-five. Both age groups of physicians practicing in the small rural counties did increase in number between 1975 and 1985. The rate of growth in the supply of young physicians, however, was faster than that of their older colleagues. Indeed, the increase in number of physicians under age sixty-five as a group was somewhat higher than those age sixty-five and over, causing a decline in the latter's relative proportion of all patient care physicians (from 18 percent in 1975 to 16 percent in 1985, despite a numerical increase of twenty-three physicians in this age bracket over the period). Examination of the age structure of individual specialties reveals a significant proportion of older physicians in general practice, a matter of future policy concern.

In the young physician (under age thirty-five) category, the rate of growth was consistent with population growth. The nonfrontier counties

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experienced a faster growth rate of young physicians than frontier areas, both in absolute numbers and as a percentage of all patient care physicians. In the category of physicians age sixty-five and older, however, while frontier counties witnessed an increase in their supply of such physicians in absolute as well as relative numbers, the nonfrontier counties saw a decline in the number of those physicians practicing there in 1985 compared to 1975.

The supply of foreign medical graduates (FMGs) increased in all categories of small rural counties over the period studied. More specifically, the number of FMGs nearly doubled between 1975 and 1985 (from 174 to 325) and grew at a faster rate than U.S. medical graduates, as evidenced by their percentages of all patient care physicians. The percentage of all patient care physicians in the United States was still higher (21 percent) than the percentage for these counties in 1985 (15 percent). Within small rural counties, frontier areas had a smaller number of FMGs in both years and witnessed a slower increase in FMG supply than did nonfrontier areas.

Trends In Geographic Variation

One of our primary objectives was to examine geographic variation in physician availability in rural counties. Exhibit 4 displays the 1985 physician availability in counties of less than 10,000 population by state. The exhibit shows only thirty-two states with counties having fewer than 10,000 residents.[8] The total number of such counties in each state is indicated.

For all thirty-two states included in Exhibit 4, the mean physician availability was 53.1 per 100,000, with 15.3 percent DOs and an average increase in availability of 14.2 percent from 1975 to 1985. The rank order of physician-to-population ratios ranged from a low of 31.3 in Georgia to a high of 86.2 in California. The percentage of DOs ranged from 0.0 percent in North Carolina and Utah to 73.7 percent in Missouri, indicating the importance of this group of physicians in the small counties of some states. The change in the ratio over the 1975–1985 decade ranged from declines of 11.4 percent in Illinois and 9.8 percent in Colorado to an increase of 85 percent in California. In interpreting the changes in availability at the state level, however, one has to keep in mind that some states have relatively small populations in small counties, and changes in several physicians can make substantial changes in the total availability and in the percentage change; for example, in California, the figures are based on only four counties with a 1985 total population of 23,200. In addition, changes in the physician-to-population ratios can be due to the

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Exhibit 4
Physician-To-Population Ratios In Small Rural Counties, By State, 1985

State[a]	Physicians per 100,000 residents	Percent DOs[b]	Percent change, 1975–1985
California (4)	86.2	5.3%	85.0%
Wyoming (8)	77.5	11.6	26.6
Kansas (61)	76.3	24.7	28.7
Washington (8)	76.0	2.6	7.6
Michigan (10)	70.5	5.6	36.9
Oregon (8)	68.0	3.3	17.4
Montana (34)	68.7	12.0	33.9
Utah (11)	65.3	0.0	20.3
South Dakota (47)	61.0	7.3	39.3
Missouri (26)	58.1	73.7	-0.3
Nevada (7)	56.6	18.2	7.0
Minnesota (12)	56.6	18.2	7.0
Oklahoma (15)	56.3	36.4	-0.3
Virginia (16)	53.6	1.7	52.3
New Mexico (9)	56.1	12.0	10.4
Iowa (15)	53.3	45.5	-4.3
Colorado (31)	53.2	3.9	-9.8
North Dakota (36)	52.8	6.5	39.3
Idaho (20)	52.0	10.5	30.3
Nebraska (61)	51.5	1.3	15.8
Texas (87)	49.7	14.8	-3.3
West Virginia (9)	49.3	4.9	3.8
Wisconsin (5)	49.2	6.7	20.0
North Carolina (8)	45.8	0.0	24.5
Florida (8)	41.9	12.0	41.1
Kentucky (24)	40.5	2.7	17.7
Mississippi (8)	35.6	11.1	1.4
Tennessee (14)	35.6	11.1	1.4
Arkansas (8)	34.6	4.3	18.9
Illinois (13)	33.3	6.7	-11.4
Indiana (5)	31.5	16.7	10.5
Georgia (47)	31.3	10.0	11.4

[a] Numbers in parentheses denote the number of counties with fewer than 10,000 residents in each state.
[b] Doctors of osteopathy.

denominator or the numerator or both; in Colorado, for example, the decline was due entirely to the increase in population with a constant number of physicians, whereas in Illinois it was a combination of both increasing population and decreasing physician supply.

To examine regional differences in physician availability in 1985, we divided the states into geographic regions using the AMA's regional aggregations (Exhibit 5). The Western states showed a mean availability of 63.9 physicians per 100,000 population; the North Central states, 58.4; and the Southern states, 43.1. The percentage of DOs ranged from 21.7

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Exhibit 5
Physician-To-Population Ratios, Percentage Of DOs, And Percentage Change In Physician Availability In Small Rural Counties, By Region, 1975–1985

Region	Number of states	Physician-to-population ratio (1985)	Percent DOs (1985)[a]	Percent change in physician-to-population ratio
North Central	11	58.4	21.7%	16.5%
East North Central	4	47.2	7.2	14.0
West North Central	7	60.2	23.5	17.9
South	11	43.1	11.2	16.0
East South Central	3	38.7	5.9	9.2
West South Central	3	49.1	18.1	5.1
South Atlantic	5	40.0	6.3	26.6
West	10	63.9	7.5	25.9
Mountain	7	60.8	8.7	14.9
Pacific	3	75.0	3.4	36.7
All counties with fewer than 10,000 residents		53.0	15.3	14.2
Entire United States		164.8	5.1	33.6

[a] Doctors of osteopathy.

percent in the North Central states to 7.5 percent in the West. The Western states showed the greatest increase in availability over the decade (25.9 percent) compared with the North Central states (16.5 percent) and the Southern states (16.0 percent).

Policy Implications

The most important policy question raised by this research is the adequacy of physician availability in the small rural counties in 1985 and in the near future. To consider this question, we must have some standard or norm of availability that is desired or optimal for these locations. To our knowledge, no such standard exists. Exhibit 6 lists several possibilities for such an estimate. The current average levels of availability in United States as a whole and in all nonmetropolitan areas are considerably higher than those in the small rural counties, but there are obvious reasons why such norms are not appropriate for smaller rural areas. The same applies to the GMENAC estimate of total physician requirements for 1990 and 2000. The figures on average primary care physician-to-population ratio (MDs only) in the entire United States and nonmetropolitan areas are presented for comparison.

Joseph P. Weiner and colleagues have recently critiqued the usefulness of a "need-based" requirements methodology such as that used by GMENAC, and present alternative approaches from health mainte-

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Exhibit 6
Alternative Estimates Of U.S. Physician Supply And Requirements

Year	Physician category (source)	Physicians per 100,000 population
1985	U.S., total patient care MDs (AMA Masterfile)	155
1985	U.S., all DOs (AOA Yearbook and Directory, 1986)	8
1985	Nonmetropolitan areas, patient care MDs (AMA Masterfile)	91
1978/2000	GMENAC "requirements," MDs and DOs (GMENAC Summary Report)	191
1985	U.S., all primary care MDs	82
1985	U.S. nonmetropolitan, all primary care MDs	50
1985	U.S. nonmetropolitan, all primary care MDs adjusted upward by 10 percent to include DOs	55
1985	HMO primary care requirements (48 percent of 120 physicians per 100,000 based on Tarlov's projection)	57.6
1985	Small rural counties, primary care MDs and DOs (current study)	40.8

nance organization (HMO) empirical practice data and HMO Delphi panels for pediatric requirements.[9] Similarly, as a starting point, we have considered Alvin Tarlov's suggestions regarding the number of physicians needed in an efficiently functioning HMO.[10] He suggests that 120 physicians would be required per 100,000 enrollees and that 55 percent of them should be primary care physicians. Adjusting for our more limited definition of primary care, we derived a target of 48 percent primary care physicians, or 57.6 per 100,000. Using this as a guideline, we see that this is almost the same as the 1985 U.S. nonmetropolitan primary care physician-to-population ratio adjusted upward by an estimated 10 percent for DOs, but is substantially higher than the 40.8 for primary care physician availability in the small rural counties we consider here. When we compared this with the primary care physician-to-population ratios for the same thirty-two states listed in Exhibit 3, we noted that only three states have availability of higher than 57.6. If the overall growth in physician supply in the next decade is similar to the 14 percent of the past decade, the availability through diffusion would reach a mean of 46.6, which is still far short of the HMO or the nonmetropolitan level; twenty-six states would still be below this level of primary care availability.

Numbers of physicians needed. It should not be concluded from this study that large numbers of physicians in aggregate may be needed in these areas. Gross calculations from these estimates using aggregate figures (that is, for all such counties as a whole) indicate that approximately 670 physicians in total would be needed to raise the current primary care physician level to the HMO standard, or 1,883 to bring

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primary care physician availability in such counties to the U.S. total availability. It is likely that the HMO standard is too stringent for application in the small rural situation, so that a figure somewhere in between the HMO and the total primary care availability may be the most appropriate. This range in additional physicians required is confirmed by results of a survey of 50 percent random sample of physicians practicing in these counties, in which 32 percent indicated that there were too few physicians in their county; their estimate of additional physicians for "adequate" availability was approximately 1,100.[11] The respondents also estimated that 74 percent of this number could be supported by their community. It is also interesting to note that the federal report cited earlier indicates that 1,148 physicians will be needed in 1994 to bring primary care physician-to-population ratios in all nonmetropolitan counties to the minimum level of primary care availability used for designation as a physician shortage area; this is a more stringent norm than any discussed previously.[12]

When we undertook this study, we wondered how much of the rapidly growing physician supply of the past decade had trickled down to the smallest of the rural communities. Our results indicate that the supply of physicians has increased in those areas but at a much slower pace, particularly in comparison with the United States as a whole or all nonmetropolitan areas during the period of this study. As to whether there is adequate physician availability in such small rural counties, we believe the answer is "no" in many of them, even though there is no definite standard of an "adequate" level of physician availability in these areas.

This article does not deal with the appropriate mechanisms for improving physician availability in such counties. Since the numbers are relatively small in the aggregate and regional differences are important, policies should allow for flexibility and creativity at the state level. Although we do not examine this question here, we believe that the federal Community Health Centers and National Health Service Corps have played important parts in the current physician availability in small rural counties, and their role in the future will be important. The South continues to be the region of greatest difficulty, and targeted efforts in this region would seem indicated.

The importance of family and general practitioners in these counties cannot be underestimated, because they provide 91 percent of primary care capability. Family practice is almost completely responsible for the rise in the board-certification percentage from 15 percent to 50 percent over the decade. The fact that most of these physicians are in general practice and that 20 percent of them are over age sixty-five indicates the

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erosion that can be expected in the immediate future as they retire. The capability of existing family practice residency programs to meet this need must be carefully examined. Likewise, the importance of DOs in many states cannot be overemphasized and must be taken into consideration in the discussions related to this issue. The relatively smaller percentage of FMGs in these areas indicates that changes in FMG policy would have less impact than in some urban areas, yet when the loss of one or two physicians can make a great difference in a given county, it cannot be ignored, either.

Future research. In the future, we intend to explore several lines of studies to carry these results further. First, we plan to do a similar analysis for all nonmetropolitan counties (MDs only) to see how other nonmetropolitan areas of various sizes compare to the counties covered in this study in their changes in physician availability. Second, we plan to perform a multivariate analysis of physician availability at the county level to see if we can determine factors that could be "predictive" of physician presence in these small rural counties. Finally, we intend to examine the frontier concept, dividing the nonmetropolitan counties into various groupings by proximity to the large urban population centers and by sociodemographic and economic characteristics, and using the Area Resource File to determine if some of the anecdotal differences between categories of rural counties can be established.

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7. We made two basic adjustments to our database. First, on physician data, since the AMA county codes and counts of physicians for those years were based on the 1970 and 1980 census and county classifications, we excluded from the AMA data those records of physicians who were practicing in counties that had experienced population increases between 1970–1975 and 1980–1985 and therefore were no longer among the counties with resident populations less than 10,000 in 1975 or 1985. This amounted to the deletion of 72 and 460 physician records (corresponding to seventeen and sixty-four counties) for 1975 and 1985, respectively. Second, because of the changing demographic status of some counties over the study period, that is, those having more than 10,000 residents in either year but not the other (the "transition" counties), we decided to concentrate on those that had a resident population under 10,000 in 1975 and 1985. As a result, 398 and 29 physician records were screened out from the 1975 and 1985 total physician counts, respectively. The constant boundary condition that emerges in this process provides a more reliable picture of the trends in the supply and availability of physicians in those counties than looking at the "artificial" boundaries that would be created if one takes the county population size alone as the criterion variable to define the boundary of interest in each year. U.S. Department of Commerce, Bureau of the Census, *Current Population Reports,* Series P-25, no. 617-698.; U.S. Department of Commerce, Bureau of the Census, *Current Population Reports,* Series P-26, no. 85-52-C; American Osteopathic Association, *1976 Yearbook and Directory of Osteopathic Physicians* (Chicago: AOA, 1976); American Osteopathic Association, *1976 Yearbook and Directory of Nonmember Osteopathic Physicians* (Chicago: AOA, 1976); American Osteopathic Association, *Yearbook and Directory of Osteopathic Physicians, 1985–1986* (Chicago: AOA, October 1985); and F.J. Popper, "The Strange Case of the Contemporary American Frontier, Or Warping the Winning of the West," unpublished paper, School of Urban and Regional Policy, Rutgers University, September 1984.
8. Several states were eliminated if they did not have counties with less than 10,000 population (eleven states), where the AMA does not report data at the county level (Alaska—judicial districts), or if there were fewer than ten physicians in all counties

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